CODED × GOODCANE · VIRTUAL PAID GIG SUBMISSION

GoodCane Weekly Pre-order

Turning TikTok interest into a weekly ordering habit, and giving GoodCane the numbers before the cane is cut.

Links

Live shop	`https://goodcane-six.vercel.app`
Staff dashboard	`https://goodcane-six.vercel.app/admin`
Demo video (2:53)	`https://goodcane-six.vercel.app/demo.mp4`
Code repository	`https://github.com/edstudios73-svg/good-cane-`
Written explanation	`github.com/edstudios73-svg/good-cane-/blob/main/docs/WRITEUP.md`
Measurement plan	`github.com/edstudios73-svg/good-cane-/blob/main/docs/MEASUREMENT.md`
This document	`https://goodcane-six.vercel.app/GoodCane-Submission.pdf`

STAFF LOGIN FOR JUDGES

Email `admin@goodcane.app` · Password `GoodCane-YmzgvnMyHB`

Try it: place an order on the shop with your own phone number, then sign in here and confirm it. You'll get a text at each step.

In one paragraph

GoodCane Weekly Pre-order turns TikTok interest into a weekly ordering habit. Customers reserve bottles for the next fresh batch, get their order number and every status update by SMS, and order the same again in one tap. GoodCane staff get a separate, protected dashboard. It shows live demand per flavour before the cane is pressed, texts customers automatically when an order is confirmed, and tracks repeat-order rate, confirmed weekly orders, cancellations and forecast accuracy. It is built with React and TypeScript on a live Supabase database, sends SMS through SasuSync from the GOOD CANE sender ID, and is deployed on Vercel. No payments are connected, and weekly reminders stay off until GoodCane approves them.

At a glance

The problem

Orders arrive as DMs and walk-ins, so GoodCane can't tell how much to press. Fresh juice that isn't sold is wasted.

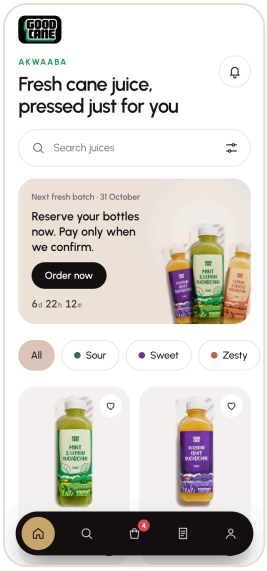
The idea

Pre-orders for each weekly batch, with a clear closing time, SMS updates and one-tap reorder.

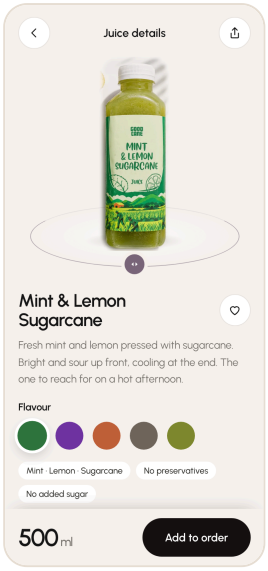
The result

Confirmed demand per flavour before production, and the four pilot measures tracked live.

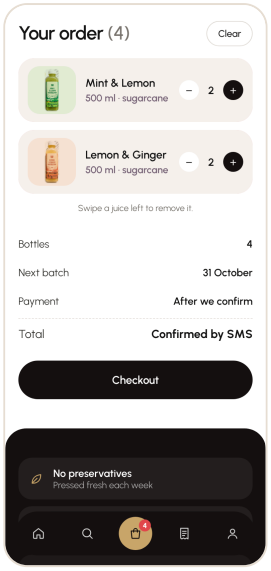
What a customer sees



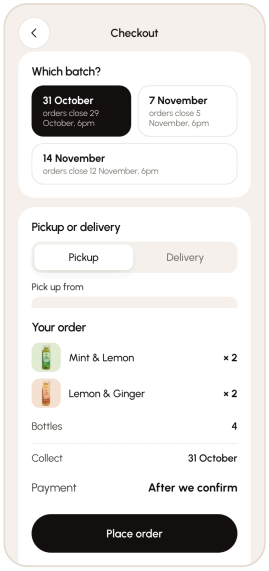
Home: next batch, countdown, live bottles reserved, taste filters



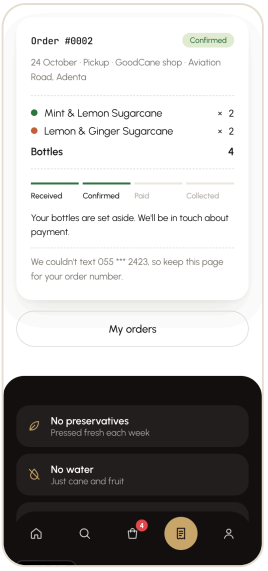
Juice page with real ingredients



Your order: nothing to pay until GoodCane confirms

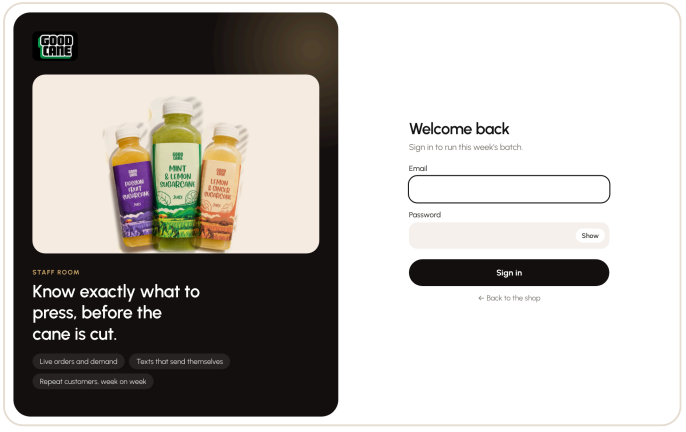


Checkout: batch, pickup or delivery, name and phone

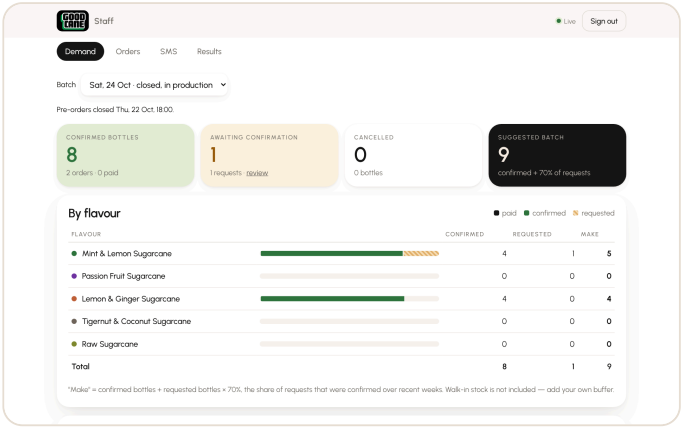


Receipt with order number and progress, also sent by SMS

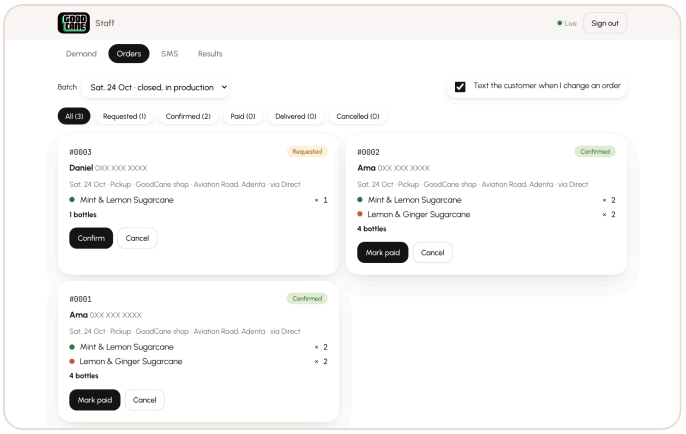
What GoodCane staff see



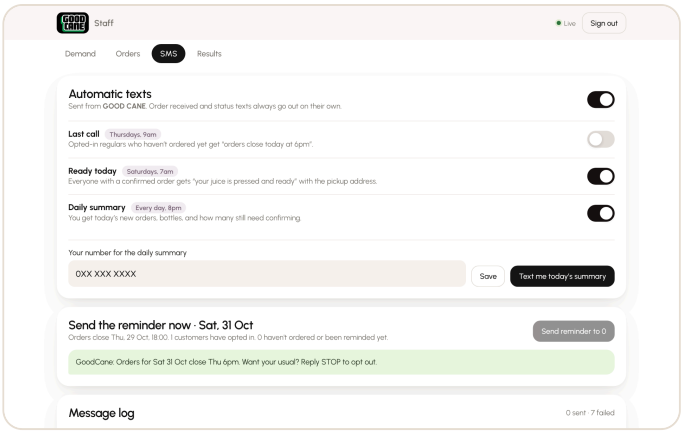
Separate, protected staff sign-in



Demand: bottles per flavour (requested, confirmed, paid) and a suggested batch size



Orders: one tap to confirm, mark paid, deliver or cancel. Each change texts the customer.



SMS: automatic texts with switches, plus every text logged with its delivery result

Phone numbers in these screenshots are masked.

Written explanation

Built for the CODED × GoodCane Virtual Paid Gig (EP11). Orders and texts are live, and the database was reset so it holds only real orders. No payments are connected; GoodCane records payment by hand. Demo video: https://goodcane-six.vercel.app/demo.mp4

1. The customer problem

GoodCane makes sugarcane juice in **weekly batches** (from its public TikTok posts: Passion fruit, Lemon ginger, Raw sugarcane, Mint lemon, Tigernut coconut). The juice has no preservatives, so it can't sit on a shelf for long. Orders currently come in through WhatsApp and walk-ins.

TikTok makes people *interested*, but interest doesn't turn into a habit:

- **For customers:** reordering takes effort. They have to remember, find the chat, retype the order, and wait to hear whether there's any juice left. Nothing prompts the second order.
- **For GoodCane:** demand only shows up as scattered chat messages. Before production day there's no reliable count of how much of each flavour people actually want. Make too much and fresh juice spoils. Make too little and sales are lost.

Problem statement: *People who discover GoodCane on social media have no easy routine for ordering again, so GoodCane can't predict weekly demand for a fresh, short-shelf-life product.*

2. Chosen approach: scheduled weekly pre-orders plus one-tap reorder

The brief offered four directions. I chose **scheduled pre-orders**, built around the batch GoodCane already makes, and added two small supports:

PIECE	WHY
Weekly batch pre-order with a cutoff (e.g. order by Thu 18:00 for Saturday)	Matches how GoodCane already produces. Demand is known *before* the cane is bought and pressed.
"Order this again" one-tap reorder	Turns order #1 into order #2. It copies the last order into the next week the customer hasn't ordered for yet.
SMS: order number, status updates and an opt-in weekly reminder	Customers always know where their order stands. Reminders are off by default, at most one a week, with a reply-STOP line. Live sending waits for GoodCane's approval.
Business demand view	Confirmed bottles vs pending requests per flavour, a suggested batch size, and the four pilot measures.

Why not subscriptions? They need fixed prices, skipping, pausing and billing rules that GoodCane hasn't set. Weekly pre-order plus reorder delivers most of the "routine" benefit and costs much less to run. Subscriptions can come later if the pilot shows strong repeat behaviour.

Why no AI? The forecast is **confirmed bottles + pending requests × the historical confirm rate**. GoodCane can read it, check it and trust it. AI doesn't solve a clear need here yet.

3. How it works

There are two separate apps built from one codebase, sharing one Supabase database:

APP	URL	WHO USES IT
Shop	`/`	Customers. Nothing on it links to the staff side.
Staff dashboard	`/admin/`	GoodCane staff, signed in with a Supabase email account listed in `staff`.

Shop (mobile-first):

1. **Discover:** links from TikTok or Instagram use `?src=tiktok`, so each order records which channel it came from.
2. **Order:**
   - Each juice has its own page.
   - Customers add bottles, then check out: pick the Saturday, pickup or delivery area, name and phone number. No payment is taken.
   - A weekly reminder text is opt-in and unticked by default.
3. **Order number by SMS:** the order is saved in the database, the order number is texted straight away, and the receipt shows the status (Awaiting confirmation → Confirmed → Paid → Collected). Customers can cancel until the cutoff.
4. **Return:**
   - On the same phone, "My orders" opens straight away.
   - On a new phone, customers sign in with a 6-digit code sent by SMS.
   - "Order this again" copies the last order into the next Saturday they haven't ordered for.

Staff dashboard:

- **Demand:** for each batch, *confirmed* bottles, bottles *awaiting confirmation*, cancellations, and a **suggested batch size per flavour**. It updates live as orders arrive.
- **Orders:** confirm, mark paid (by hand, since payment happens outside the app), mark delivered, or cancel with a reason. Each change texts the customer, which can be switched off per session.
- **SMS:** send the weekly reminder to opted-in customers who haven't ordered yet, and see every text with its delivery result from SasuSync.
- **Results:** repeat-order rate, confirmed orders per week, cancellation rate and forecast accuracy.

SMS

TEXT	WHEN IT'S SENT
Order received + order number	Customer places an order
Confirmed / Payment received / Delivered / Cancelled	Staff change the status, or the customer cancels
Weekly reminder (opt-in only, with STOP line)	Staff press "Send reminder" before the cutoff, or automatically on Thursday 9am once GoodCane switches "Last call" on

TEXT	WHEN IT'S SENT
Ready today	Automatically at 7am on Saturday, for every confirmed order
Daily summary to staff	Automatically at 8pm: new orders today, bottles confirmed for Saturday, orders still to confirm
Sign-in code	Customer asks to see their orders on a new phone

How sending works:

- Texts are written by `public.sms_text()` from fixed templates, all under 160 characters, so nobody can make the system send arbitrary messages.
- They are sent from the database through SasuSync using `pg_net`. The API key is stored in Supabase Vault, not in the website.
- Delivery results are read back into `sms_log` every minute.
- Scheduled texts run with `pg_cron` in Accra time (UTC). Each has its own switch in SMS → Automatic texts. The Thursday last call ships switched **off**, because reminders need GoodCane's approval.
- Sending is controlled by the `sms_live` setting and can be paused instantly.

Security and data

- **Customers can't read tables.** Every customer action is a database function that checks its input: phone format, open batch, known juices and locations, quantity limits, and a rate limit per number.
- **Proving who you are:**
  - The phone that placed an order gets a private token.
  - Another phone needs the SMS code. Codes are hashed, expire after 10 minutes and allow 5 attempts.
  - Ordering with someone else's number doesn't reveal their orders.
- **Staff access:** staff read data through row-level security and change orders only through `staff_*` functions that enforce the allowed status steps.

Requests vs confirmed vs paid

A request is **not** a sale. Only `confirmed`, `paid` and `delivered` count as firm demand. Requests are shown separately and weighted in the forecast by how often requests have historically been confirmed. "Paid" is set by GoodCane by hand because no payment integration exists.

Privacy

- **Data collected:** name, phone number, flavour quantities, pickup point or delivery *area*, and an optional note. No exact address is stored; GoodCane asks for it by phone when delivering.
- **Texts:**
  - Order updates go to the number the customer gave for that order.
  - Weekly reminders are opt-in, at most one a week, and can be stopped in My orders or by replying STOP.
- **Remove my details:** wipes the customer's name and number, cancels their unpaid open orders, and keeps anonymous totals for forecasting.

- **Sample history (optional):** `supabase/sample-data.sql` adds six weeks of labelled history (`is_sample`) for testing the Results tab. Those rows are never texted, and `supabase/reset.sql` clears everything for a fresh start. The live demo runs with real orders only.

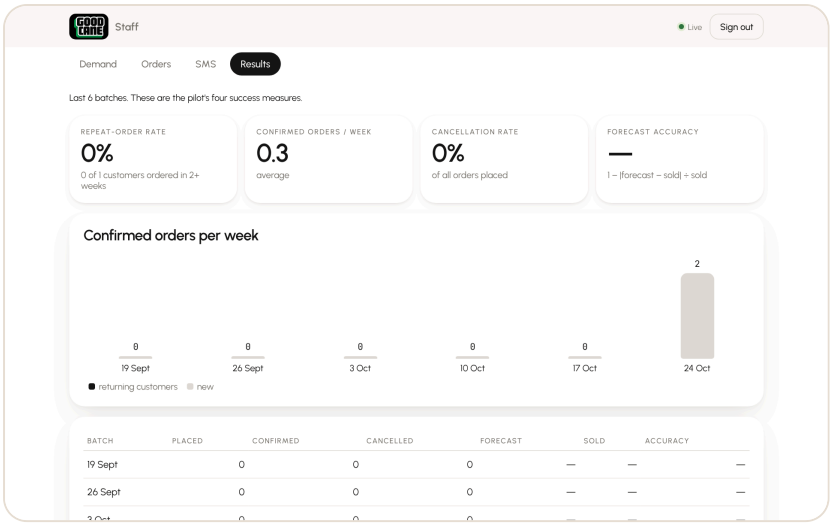
4. Assumptions to confirm with GoodCane

#	ASSUMPTION	WHY IT MATTERS
A1	Production is **weekly**, on a fixed day (currently: Saturday)	The whole schedule is built around it
A2	Pre-orders can close **~2 days before** production (currently: Thu 18:00)	Time needed to buy cane and plan the press
A3	Flavour line-up matches the public posts and is stable week to week	The menu
A4	Bottle size(s) and **prices**: unknown, so not shown	Totals and payment
A5	Pickup points and **delivery areas/fees**: unknown, so placeholders are used	Fulfilment options
A6	Payment is by mobile money or on collection, confirmed by GoodCane by hand	The "Paid" step
A7	Customers are happy to get order updates and an opt-in weekly reminder by **SMS** from a "GoodCane" sender ID	Reminder channel; sender ID must be registered with SasuSync
A8	Shelf life is short enough that over-production means waste	The value of forecasting
A9	Someone at GoodCane can check the dashboard daily during the pre-order window	Confirmations happen in time
A10	Walk-in sales continue alongside pre-orders	The forecast covers pre-orders only; GoodCane adds a walk-in buffer

5. Rollout plan

PHASE	WHEN	WHAT
0. Validate	Week 0	Walk through the live site with GoodCane. Confirm A1–A10. Set real flavours, sizes, prices and pickup/delivery areas. Agree SMS wording and approve live sending.
1. Hand over	Week 1	Give GoodCane staff their logins. Register the GoodCane sender ID with SasuSync under GoodCane's account, agree the text wording, set real sizes, prices and pickup points, and delete the sample history.
2. Pilot	Weeks 2–5	20–30 existing WhatsApp customers who opt in. Share the link on WhatsApp Status and in the TikTok bio with `?src=` tags. Texts run on GoodCane's approved sender ID; reminders go out only when staff press Send, or on Thursday once GoodCane turns on the automatic last call.
3. Review	Week 6	Compare against the success thresholds in MEASUREMENT.md. Decide whether to open to everyone, add mobile-money payment (with GoodCane's approval), or try subscriptions.

Measurement and pilot plan



Results tab: the four measures, live

The staff dashboard calculates all four measures in the **Results** tab (`/admin/`). Until real orders build up, the numbers there come from labelled sample history.

The four measures

MEASURE	DEFINITION	WHERE IT COMES FROM	PILOT TARGET (TO AGREE WITH GOODCANE)
Repeat-order rate	Of customers with ≥1 confirmed order in the period, the share with confirmed orders in **2+ different weeks**	Orders table (status ∈ confirmed, paid, delivered)	≥ 40% over 4 weeks
Confirmed weekly orders	Number of orders per batch that GoodCane confirmed (confirmed, paid or delivered)	Orders table, grouped by batch	Week 4 at least 1.5× week 1
Cancellation rate	Cancelled orders ÷ all orders placed (split by who cancelled: customer or business)	Status history	≤ 10%; business-side "sold out" cancellations tracked separately
Forecast accuracy	`1 - |forecast - sold| ÷ sold` per batch, averaged (equivalent to 1 – absolute percentage error)	Forecast saved at cutoff + bottles sold entered after batch day	≥ 85% by week 4

Supporting signals, all already captured in the data:

- **Source of first orders:** TikTok, Instagram, WhatsApp or direct, from `?src=`. Shows which channel turns discovery into orders.
- **Reorder share:** orders created through "Order this again" or a reminder text (`source = reorder | reminder`).
- **Reminder opt-in rate and opt-outs:** whether customers want reminders at all.
- **Request → confirmed rate:** feeds the forecast, and shows whether GoodCane can confirm everything people ask for.

- **Waste:** bottles made – bottles sold, the real cost the forecast aims to reduce.

How forecast accuracy is captured

1. At cutoff, the Demand tab shows a **suggested batch** (confirmed + pending × confirm rate). Staff tap **Save forecast**.
2. After batch day, staff enter **bottles made** and **bottles sold**.
3. Results shows accuracy per batch and on average.

Pre-orders don't cover walk-ins, so we also track *pre-order share of sales*, i.e. pre-ordered bottles ÷ sold. As that share grows, the forecast covers more of real demand.

Small pilot to validate demand (4 weeks)

Who: 20–30 people who have already bought from GoodCane on WhatsApp and agree to take part. TikTok followers can join through a tagged bio link.

Baseline (week 0): record each customer's orders over the past 4 weeks from GoodCane's WhatsApp history and sales notes. This gives a "before" repeat rate and a weekly order count. Also record bottles made vs sold for the last 4 batches.

Weekly routine:

DAY	ACTION
Fri–Thu	Pre-orders open for Saturday's batch
Wed	GoodCane approves the reminder text; staff send it to opted-in customers who haven't ordered
Thu 18:00	Cutoff. Staff confirm or cancel requests, then **save the forecast**
Sat	Produce, fulfil, mark paid and delivered
Sun	Enter bottles made and sold. Five-minute review of the Results tab

What counts as validated demand:

- At least 60% of pilot customers place one or more pre-orders.
- Repeat-order rate of 40% or higher (vs the WhatsApp baseline).
- Forecast accuracy of 85% or higher in weeks 3–4, with less waste than baseline.
- Cancellation rate of 10% or lower.

What we'd learn even if targets are missed:

- **Low pre-orders but high walk-ins:** people prefer spontaneous buying, so focus on stock alerts instead.
- **High cancellations:** the cutoff is too early or the confirmation is too slow.
- **Low reminder opt-in:** remove reminders and rely on "Order this again" and WhatsApp Status.

Feedback: a 3-question WhatsApp check-in at week 2 and week 4:

1. Was ordering easy?

2. Would you keep pre-ordering?
3. What almost stopped you?

GoodCane approves it before it's sent.